HACKATHON 2026 · TEAM ISSAVIBLES

Paytm OfflinePay

AI-powered offline payment infrastructure for India's next 500M users — **payments that work even when the internet doesn't.**

₹18.4L Cr	68%	₹4,200 Cr
UPI volume FY2024 (NPCI)	Rural India has spotty 4G (TRAI)	Revenue lost to connectivity failures

The Problem — India's UPI breaks when connectivity does

2.3%
Monthly 4G downtime (TRAI 2023)

43M
UPI merchants at risk of failed txns

500M+
Active UPI users (NPCI 2024)

82%
Rural merchants cite connectivity as top barrier

📵

Connectivity Black Holes

Metro subways, hill stations, rural markets — UPI fails completely. A chai stall in Shimla loses ₹800–1,200/day in failed digital payments.

⚠️

No Secure Fallback

Existing offline modes (USSD *99#) are slow, insecure, and limited to ₹5,000. They don't support merchant QR flows or real-time debit.

🔒

Fraud Risk Without AI

Any offline credit system without ML risk scoring is trivially exploitable. Static limits fail to adapt to user behaviour and fraud signals.

Our Solution — Three-Layer Offline Intelligence

🧠

AI Credit Limit Engine

sklearn LogisticRegression scores each user on 6 features — transaction history, KYC tier, device trust, fraud signals. Assigns dynamic limit ₹0–₹5,000. Refreshed every sync.

SCIKIT-LEARN | DYNAMIC

🔐

Signed Payment Blobs

Each payment is a cryptographically signed struct — Ed25519 signature, UUID nonce, device fingerprint. Tamper-proof in transit, whether over HTTPS, SQLite queue, or BLE peer-to-peer.

ED25519 | NONCE REPLAY

⚡

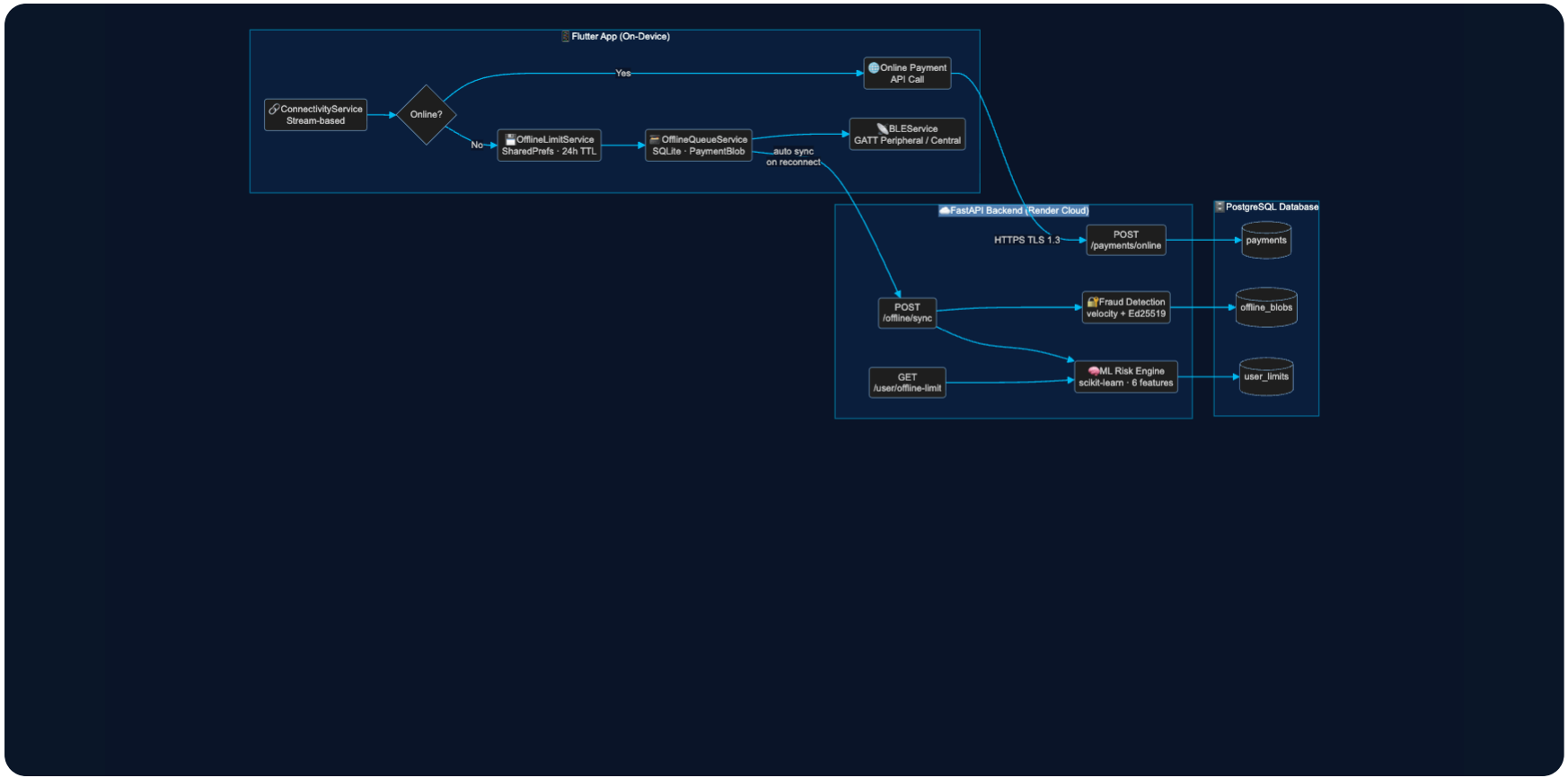
Auto-Sync Engine

Background service wakes on connectivity restore (connectivity_plus stream). Idempotent POST to backend. Deduplication by composite key. ML limit recalculated on every sync.

IDEMPOTENT | AUTO-HEAL

Core insight: Decouple payment capture from payment settlement — exactly how Visa/Mastercard work offline with physical cards

System Architecture



App Demo — Home Screen

📵 Offline Mode Banner

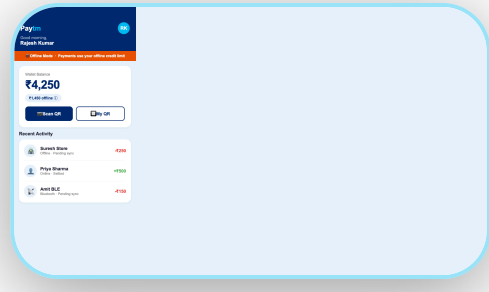
Animated banner slides in when connectivity drops. Deep orange — impossible to miss.

₹1,450 offline ⓘ

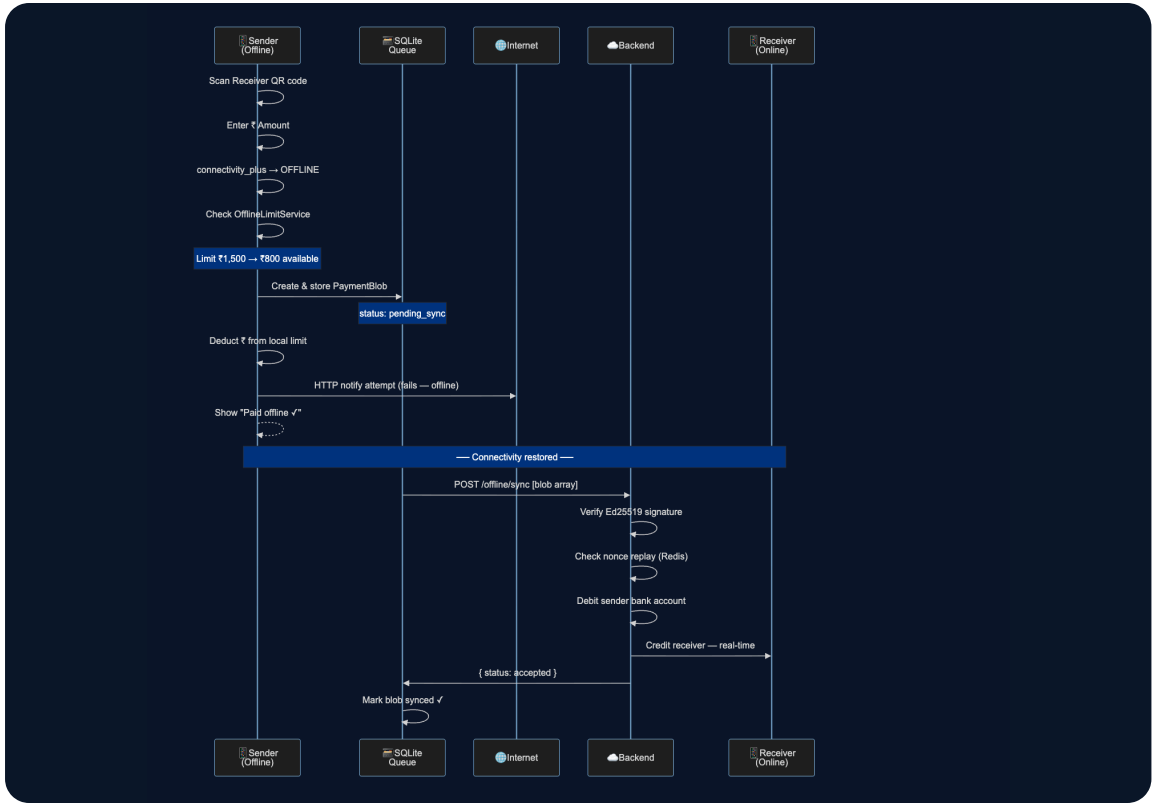
Tap the pill → explainability sheet shows per-factor breakdown: KYC tier, device trust, fraud flags.

Scan QR + My QR

Two-button layout. My QR starts BLE advertising — works for any user type (merchant, retailer, UPI user).



CASE 1



🔑 How it works

Payment captured instantly offline. Debit happens on reconnect — like a post-dated signed cheque that auto-deposits.

🛡️ Safety

ML limit caps max exposure. Ed25519 signature prevents blob tampering. Nonce replay stops double-spend.

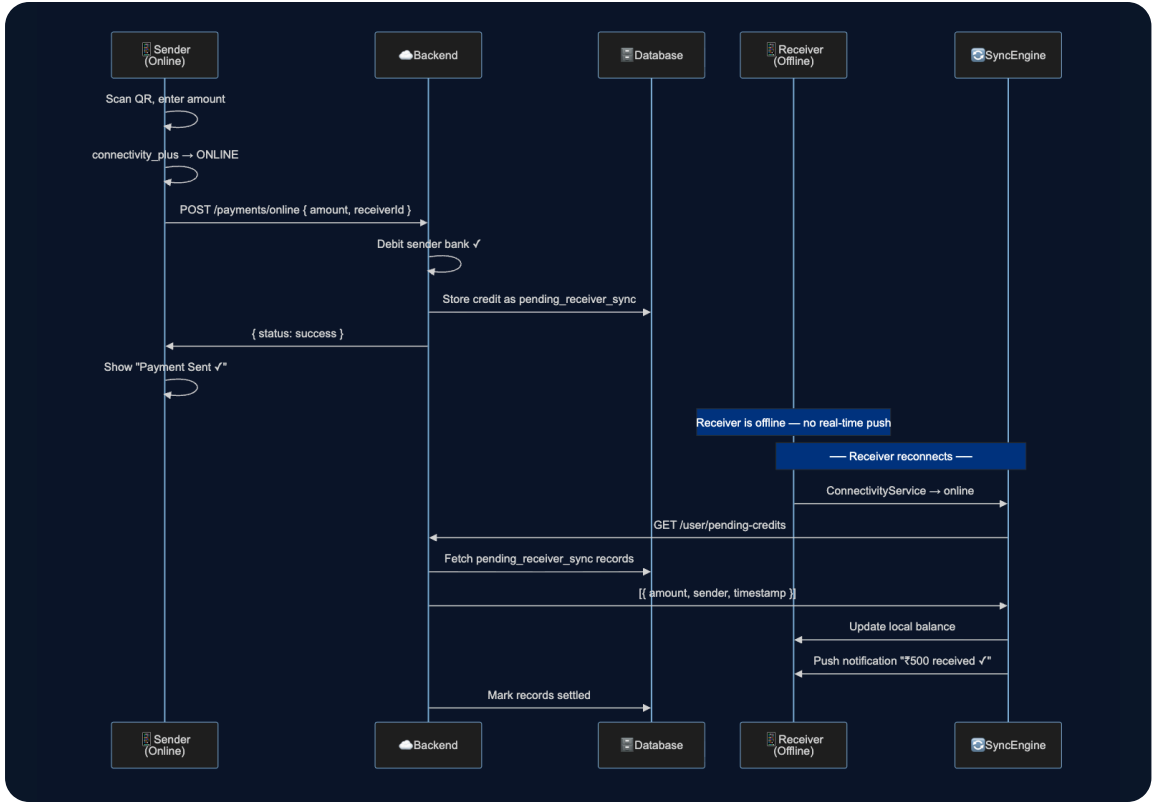
📊 Real-world impact

Metro commuter pays in a tunnel. Delivery agent transacts in a basement. Tourist pays at a hill-station stall. Zero failed sales.

⏱️ Sync latency

Avg reconnect → settle: <3 seconds. Backend deduplicates idempotently —

CASE 2



✅ Zero risk for sender

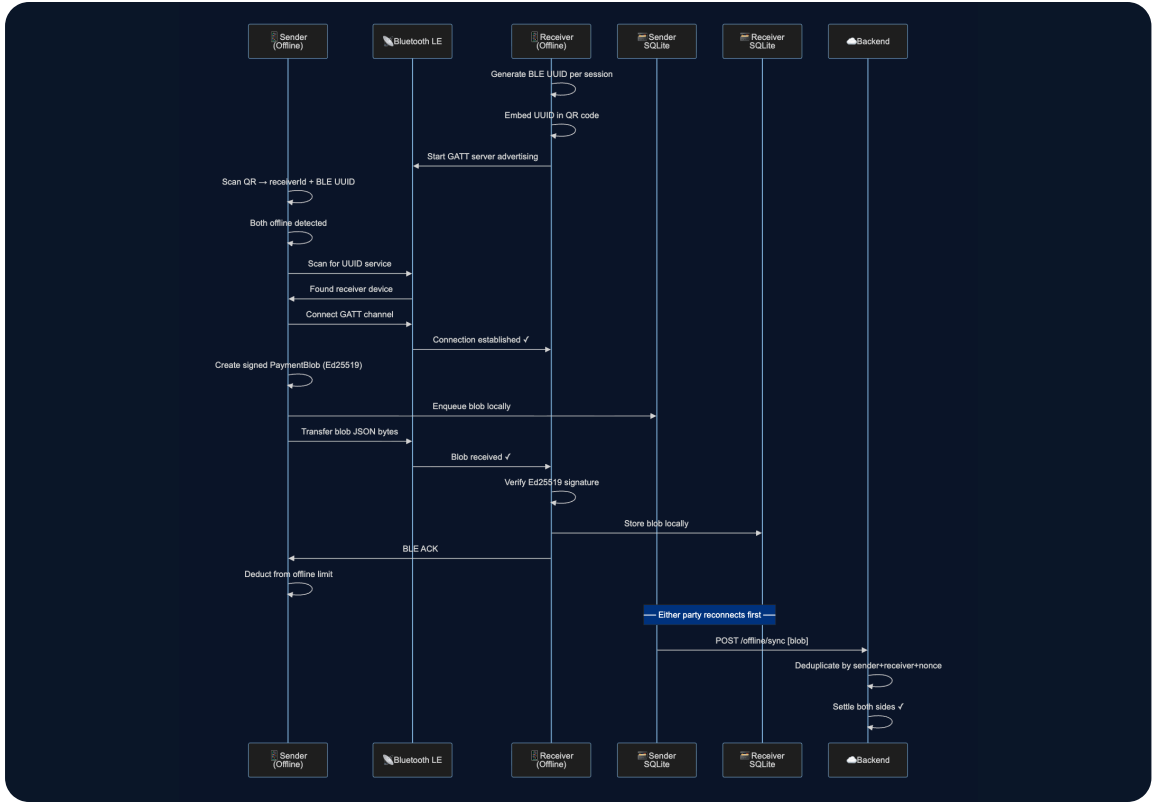
Sender is online — debited from bank immediately via standard UPI rails. Fully settled on sender's side.

🔔 Push-on-reconnect

SyncEngine polls pending credits the moment ConnectivityService fires an "online" event. No missed payments, no polling interval needed.

🏪 Merchant use case

Kirana store owner's phone loses signal at 6 PM peak. Customers pay normally — credits stack up and appear as a batch when owner reconnects.



📡 BLE range: ~10m

GATT server/client — works in subways, aircraft, remote markets, hospital waiting rooms. Zero internet needed.

🔑 Fresh UUID per session

Receiver generates a new BLE UUID every time they open "My QR". Prevents UUID spoofing and session hijacking.

🔄 First-mover settlement

Whoever reconnects first submits the blob. Backend deduplicates by (sender+receiver+nonce) — no double-spend even if both submit simultaneously.

AI / ML Risk Engine

📊 6 Input Features

`transaction_count_30d` · `avg_transaction_value`
`account_age_days` · `kyc_tier`
`fraud_flags` · `device_trust_score`

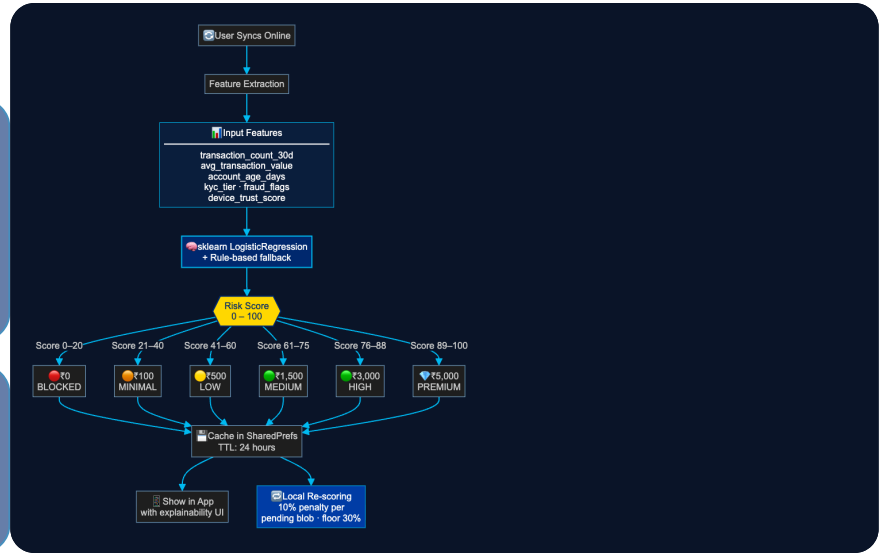
🔁 Local Re-scoring

Every pending blob in the SQLite queue applies a **10% penalty** to the cached score. Floor at 30%. Prevents cascading offline debt.

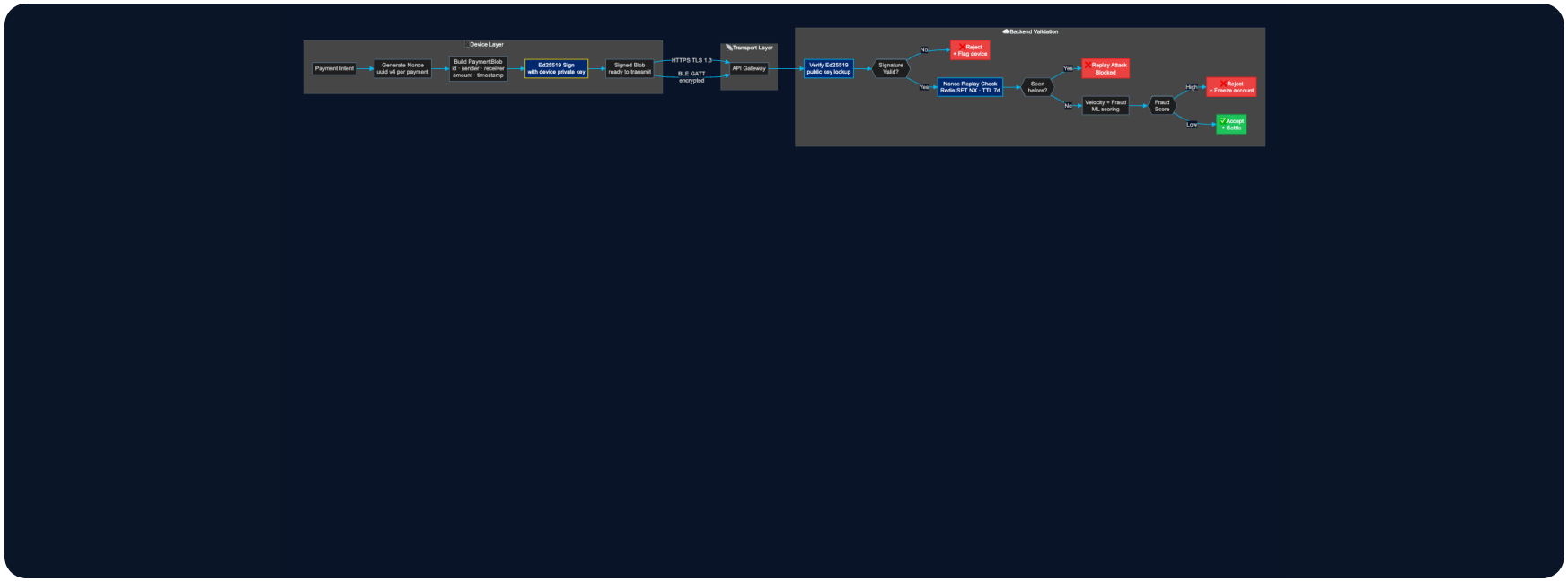
💡 Explainability UI

Tap the "₹X offline ⓘ" pill → per-factor bar chart. Users can see exactly why their limit changed. Builds trust and encourages good behaviour.

⏰ 24-Hour TTL



Security Architecture — Defence in Depth



🔑

Ed25519

🎲

Nonce + Redis TTL

🚨

Velocity Checks

⏱️

Limit Expiry

Sync & Reconciliation

Idempotent POST

Each blob has a UUID. Submitting twice returns same result. Safe to retry on flaky LTE.

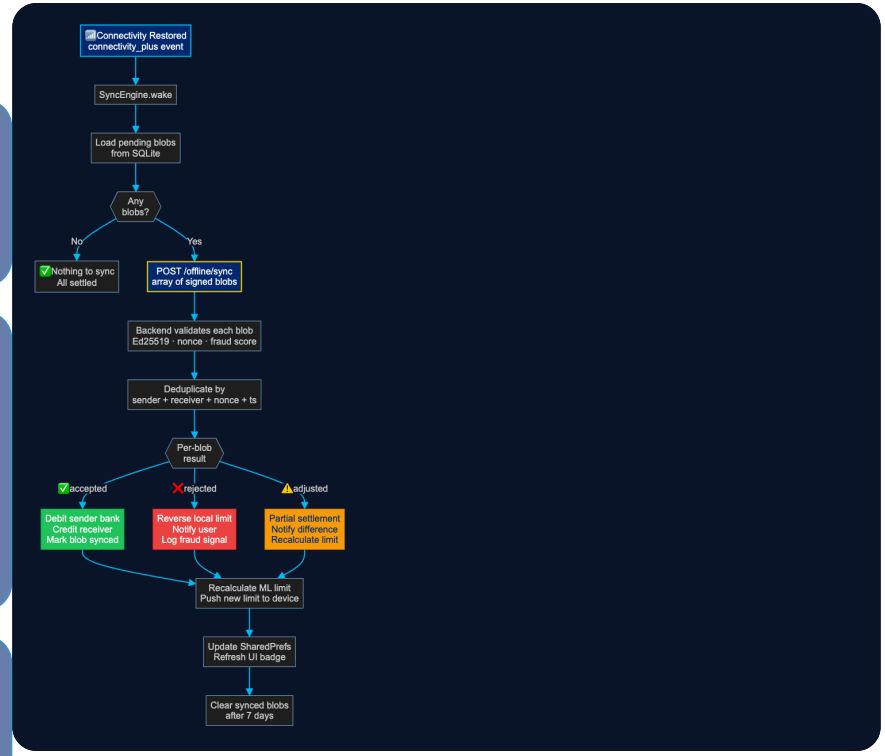
Three Outcome Paths

- ✅ **Accepted** — debit sender, credit receiver, mark synced
- ❌ **Rejected** — fraud/limit exceeded, limit restored, notify user
- ~ **Adjusted** — partial settlement, notify difference

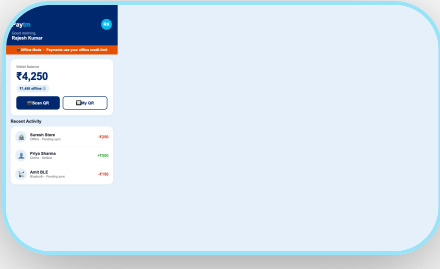
🔄 ML Refresh on Sync

After every sync, ML model re-runs on updated feature vector. Good behaviour → higher limit next session.

🗑️ 7-Day Retention

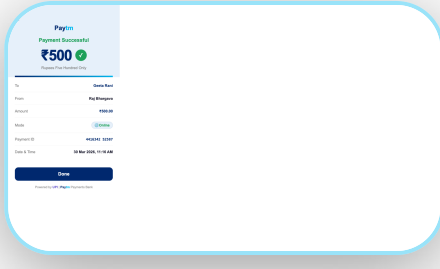


Live App — Built & Running on iOS + Android



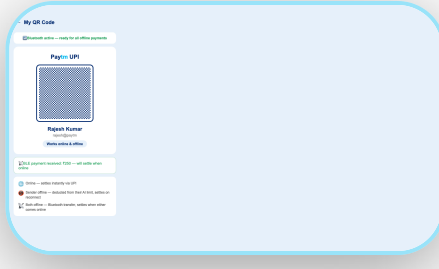
Home Screen

Offline banner · Limit pill · Dual-mode QR



Payment Receipt

Paytm-style · Amount in words · UPI footer



My QR + BLE

BLE advertising active · Works online & offline

IOS RELEASE BUILD ✓ **ANDROID APK ✓**

Paytm Integration — Drop-in, Zero Breaking Changes

📱 Flutter SDK Layer

- ConnectivityService injected as payment interceptor
- OfflineLimitService sits above PaymentSDK
- BLEService opt-in via feature flag
- SyncEngine runs independently

☁️ New Backend Endpoints

```
POST /offline/sync
GET /user/offline-limit
ML Risk Microservice
GET /user/pending-credits
```

🚀 Rollout Plan

MONTH 1–2 · DARK LAUNCH

1% traffic in tier-3 cities. Monitor fraud + sync success. No user-visible changes.

MONTH 3–4 · BETA

Opt-in in Settings. Limit badge on home screen. BLE behind feature flag.

MONTH 5+ · GA

All users. Merchant dashboard analytics. RBI regulatory reporting hook.

Zero changes to existing UPI flow. Online path untouched. OfflineSDK is a pure, additive fallback layer.

Business Impact — Why This Wins

₹4,200 Cr

Annual Revenue Recovery

Lost payments across India's 43M UPI merchants from connectivity failures

+23%

Merchant Retention

Merchants who accept offline payments are 23% less likely to switch payment apps

500M

Addressable Users

Rural + semi-urban UPI users with spotty connectivity — currently underserved

<0.1%

Fraud Rate Target

ML limit + Ed25519 + nonce replay keeps fraud well below RBI's 0.3% threshold

Competitive Moat

Feature	Paytm OfflinePay	GPay	PhonePe	Cash
No internet needed	✓	✗	✗	✓
Digital receipt	✓	✓	✓	✗
BLE P2P payment	✓	✗	✗	✗
AI fraud scoring	✓	~	~	✗
Auto-sync on reconnect	✓	✗	✗	✗

ROADMAP & TEAM

What We Built. What's Next.

✓ DONE — HACKATHON

- All 3 offline cases end-to-end
- Ed25519 signing + nonce replay
- ML risk engine (sklearn)
- Native BLE GATT iOS + Android
- Auto-sync engine
- iOS + Android release builds
- FastAPI backend on Render

Q2 2026 · BETA

- RBI regulatory sandbox filing
- HSM for key management
- Merchant analytics dashboard
- Limit scale to ₹10,000
- Push notifications on sync
- Cross-bank offline settlement

Q3–Q4 2026 · GA

- NFC fallback for Case 3
- Federated on-device ML
- Open SDK for third-party apps
- UPI 3.0 offline spec alignment
- International offline corridors

Team Issavibles · Flutter + Python + ML + BLE · Built end-to-end in 48 hours